

TANISHQ SOMANI

M.S. Computer Science – available now, open to relocation • (865) 684-5948 • tsomani@vols.utk.edu • [GitHub](#) • [Portfolio](#) • [LinkedIn](#)

Full-stack engineer (M.S. Computer Science, 2026; B.S. Microbiology) who ships real products end to end. Built and operate a live consumer app with real users, built a deployed clinical AI pipeline improved from 53% to 86% accuracy through evaluation-first iteration, and contributed an open-source fix to uv, a widely used Python tool, in Rust, a language I taught myself for the job. Professional experience building data pipelines, integrations, and workflow automation. I ran my own business for three years before CS, so I default to ownership, and I build with AI tooling (Claude Code) daily.

WHAT I'VE BUILT

[Que – Production Full-Stack App \(Live, Real Users\)](#) | TypeScript, React, Next.js, Node.js, PostgreSQL, Redis

A fitness tracking app with social challenges and an in-app coin economy, live in production with real users.

- Designed, built, shipped, and operate it solo: REST APIs, PostgreSQL schema, a Redis-locked background job engine, and full observability (Sentry error tracking, typed event telemetry).
- Engineered idempotent, transaction-safe settlement (database transactions + compare-and-set guards) so a retry or replay can never double-pay, and a multi-device sync engine with per-day conflict resolution that never silently drops an edit.

[DxMap – Clinical NLP Pipeline: Unstructured Notes → Standardized Codes](#) | Python, FastAPI, LLM APIs, RAG

An AI tool that reads a doctor's free-text note and returns standardized ICD-10 billing codes with confidence scores and rationale.

- Improved top-1 accuracy from 53% to 86% by building an evaluation harness first and fixing measured failure modes across a ~74,000-entity corpus; output includes a human-review flag for low-confidence extractions.
- Designed it as composable stages (negation handling, query decomposition, hybrid BM25 + dense retrieval with reciprocal rank fusion, LLM reranking across OpenAI/Anthropic/Gemini, isotonic confidence calibration).

[astral-sh/uv – Open-Source Fix in Rust \(Self-Taught\)](#) | Rust | PR #19762 (under review) | CodePath AI301 capstone

A bug fix contributed to uv, Astral's widely used Python package manager.

- Found a bug that crashed every uv command, taught myself enough Rust to trace the real root cause across crate boundaries when my first assumption was wrong, and shipped the fix with integration tests.

EXPERIENCE

Software / Automation Developer (Graduate Assistant)

Jan 2025 – May 2026

University of Tennessee, Office of Innovative Technologies – Knoxville, TN

- Built data pipelines and integrations (JavaScript, SQL, webhooks, APIs) that ingest, structure, and route 450+ requests annually across disparate systems, including standardizing a records-request process that saves ~1 hour per request.
- Reverse-engineered an undocumented legacy system and rebuilt it 0 to 1 into reliable, composable workflows in production; drove rollout and authored the handoff documentation the next team still runs on.

AI Tooling Instructor – AKDN

Nov 2024 – Mar 2025

Volunteer youth education program – Remote

- Led a remote class teaching high-school students modern AI tools, mentoring them through an independent AI project from concept to completion.

Co-Founder & Operator – Ionian Bulk

2020 – 2023

Multi-category resale business

- Co-founded and ran a resale business spanning sneakers, apparel, GPUs, and gaming cards – sourcing inventory, operating vendor booths at major national events, and growing an Instagram audience to 8,000+ followers.

Store Manager — City Park Market (concurrent with Ionian Bulk)

2020 – 2024

EDUCATION

University of Tennessee, Knoxville – M.S., Computer Science

Aug 2024 – May 2026

GPA 3.6 / 4.0 • Coursework: NLP, Advanced Software Engineering, Algorithms • CodePath Advanced AI (AI301), in progress

East Tennessee State University – B.S., Microbiology

Aug 2015 – May 2020

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust (open source), Java, C++

Engineering: Backend services, data pipelines & integrations, REST APIs, FastAPI, Node.js, Next.js, React, React Native, PostgreSQL, Redis, webhooks, cron/background jobs, Docker, Git, CI/CD, unit & integration testing, observability (Sentry, telemetry), Linux

AI/ML: LLM APIs (OpenAI, Anthropic, Gemini), agentic pipelines, RAG & hybrid retrieval (RRF), evaluation harness design, confidence calibration, prompt engineering; Claude Code daily

Endurance athlete: 3x marathon & 2x half-marathon finisher; training for a full Ironman.